



5G at World Cup

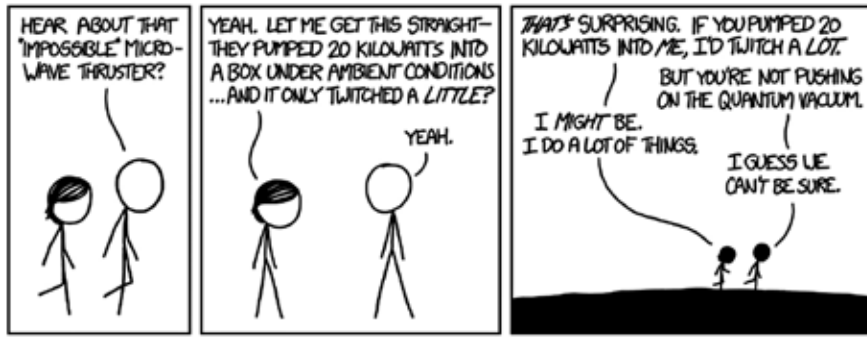
Huawei is planning to test a prototype network of 5G along with Russian carrier MegaFon in the FIFA World Cup 2018 <http://dgit.in/Huawei5G>



Moto KeyLink

Motorola's KeyLink finds your lost smartphone and keys through bluetooth, from as far as 100 feet away <http://dgit.in/MotoKeyLink>

Image: <http://kcd.com/1404/>



by expelling a part of its mass with high speed. This expelled part is the fuel, which is combusted to produce thrust. However, the irony of the equation is that the greater the speed required, the greater the fuel mass that needs to burn, further increasing the total mass of the craft – this problem is affectionately called 'The Tyranny of the Rocket Equation'.

The Rocket Equation isn't flexible, and depending on the destination, usually results in 83-96 percent of the craft mass being composed of fuel. That's the cost of space travel. And even with fuels with the highest energy content in use today, such as hydrogen-oxygen mixes, we still spend a considerable amount (83 percent) of craft mass as fuel in getting enough thrust to escape gravity. Keeping this reality in mind, the potential benefits of a fuel-free propulsion system are the stuff that dreams are made of. It would allow us immeasurable flexibility in space flight at very little cost. Such a system would be akin to what we see in Science Fiction, where space crafts silently and smoothly take off and land from planets, and are able to freely manoeuvre along the emptiness of space without having to factor in precise trajectories to conserve fuel.

First Mars, then the universe!

If scientists are able to reconcile and explain the "anomalous thrust" generated from EmDrive devices, Cannae drives and any other seemingly quantum-based reactionless propulsion, we may have to rewrite some physics textbooks. But, it would also mean that we've finally crossed the immense hurdle posed in our exploration of space and everything that comes along with it. By solving

the energy problem in space travel, we can envision a future where potential colonisation and habitation efforts can be affordably designed.

Reactionless drives don't necessarily mean free fuel, since electricity will always be the driving force – at least under current designs. Their greatest immediate utility won't even be to get a spaceship off the surface of Earth, but to make interstellar space travel faster and cheaper. Using the energy from these new drives, NASA

would be in addition to the hundreds of tonnes of fuel required to get us into orbit. The journey would further be dependent on factors such as planetary alignment for fuel-efficient travel trajectories. This would mean that a human crew would need to stay near or around Mars for nearly 18 months in wait for their return launch window, which means additional supplies and mass to the spacecraft. With the new reactionless drives, this would be reduced to a couple of hundred tonnes of combined mass with a total of eight months travel which includes a two-month stay on Mars. With improved technology over time, we can easily imagine this ideal future.

Sceptic but hopeful

Good science is sceptical by nature. It questions and critiques every aspect of claims made by eager researchers. So, even as the results of NASA's experiments face closer scrutiny and criticisms, we know that in the long run truth will be discovered. And even as we hope that



If proven, reactionless drives could support a generation of space travel for everyone

conservatively estimates that we can feasibly conduct a manned mission to Mars from Earth orbit at significantly cheaper and faster rates. With the help of the new drives, a 90-tonne spacecraft could use a 2-megawatt nuclear-powered reactor to generate 800 newtons of thrust.

Using current technology, there would be thousands of pounds of fuel required for a trip of this scale and would take six months each to travel back and forth. This

scientists will discover new methods and technologies to make the EmDrive and Cannae drives possible, we know that even in failure it isn't the end of innovation. Other means of achieving the same goal such the theoretical 'Alcubierre' drive and solar sails are also being explored. Physics may have its limits that stop reactionless drives from becoming a reality, but there's no limit to the human capacity for imagination and innovation. d